LATENT-SPACE DENSITY MODELING FOR RESOLVABLE-SOURCE DETECTION IN CONFUSION-LIMITED LISA SIMULATIONS

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INTRODUCTION

LISA will operate in a confusion-limited regime where unresolved Galactic binaries form a structured foreground across much of the millihertz band [1]. In this setting, the problem is not simply detecting weak signals in instrumental noise, but distinguishing resolvable sources embedded within overlapping foreground structure. Here we study one-class detection using continuous wavelet transform (CWT) representations and latent embeddings learned by a convolutional autoencoder trained only on foreground segments [2]. We compare local geometric deviation in latent space against explicit latent density modeling under a fixed preprocessing and data-generation pipeline [3].

DATA GENERATION

Confusion foreground = Instrumental noise + unresolved Galactic binaries

Positive samples = foreground + injected resolvable source (MBHB, EMRI, bright GB)

Segment duration = 3600 s sampled at 1 Hz

Training/Test set = 5000 foreground segments / 200 foreground + 400 signal segments

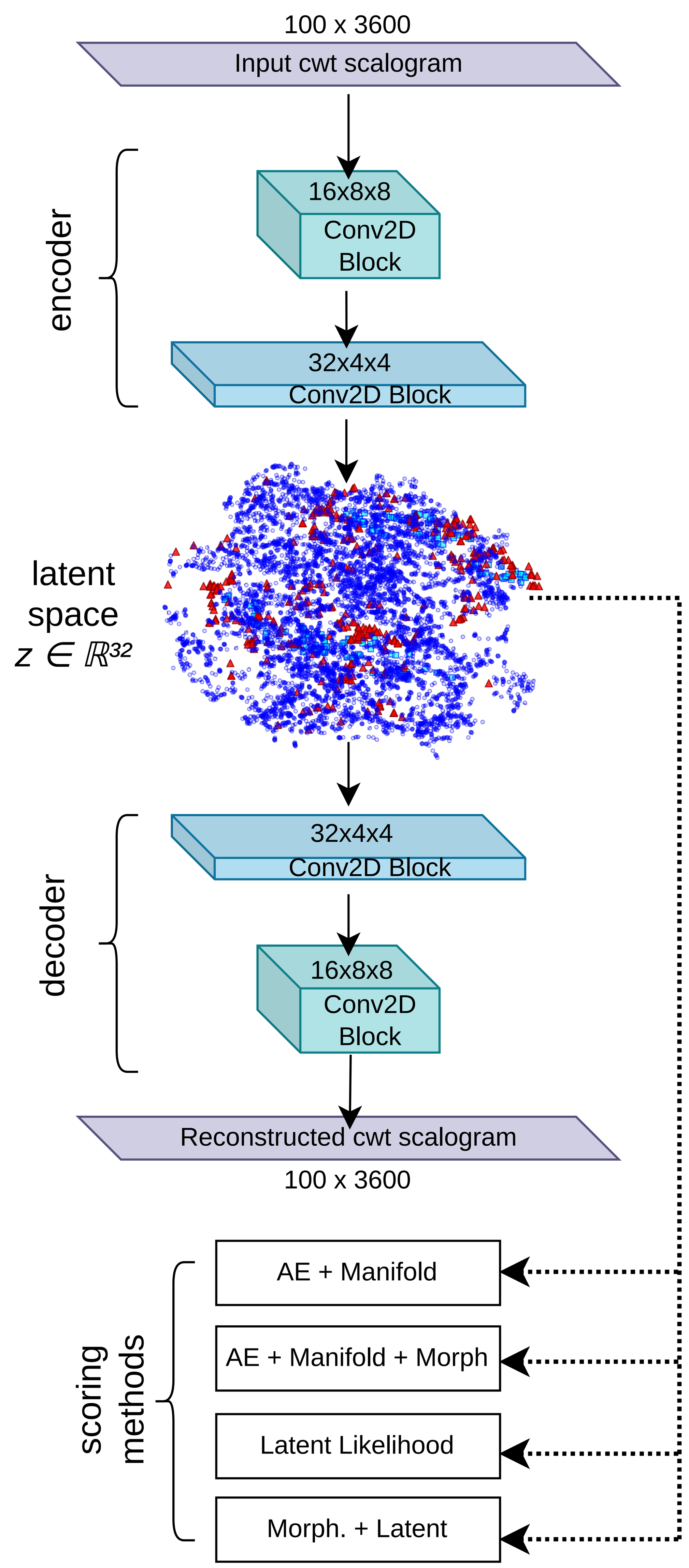
Representation = CWT scalograms

SCORING METHODS

AE+Manifold*	$\alpha e_{\mathrm{AE}} + \beta d_{\mathcal{M}}$
AE+Manifold+Morph*	$(1-\lambda)s + \lambda d_{\phi}$
Latent Likelihood**	$-\log p(f_\theta(x))$
Morph+Latent**	$-\log p((\phi(x), f_\theta(x)))$

*Local latent deviation.
**GMM or KDE estimators.

LATENT SPACE FRAMEWORK



PRIMARY RESULTS

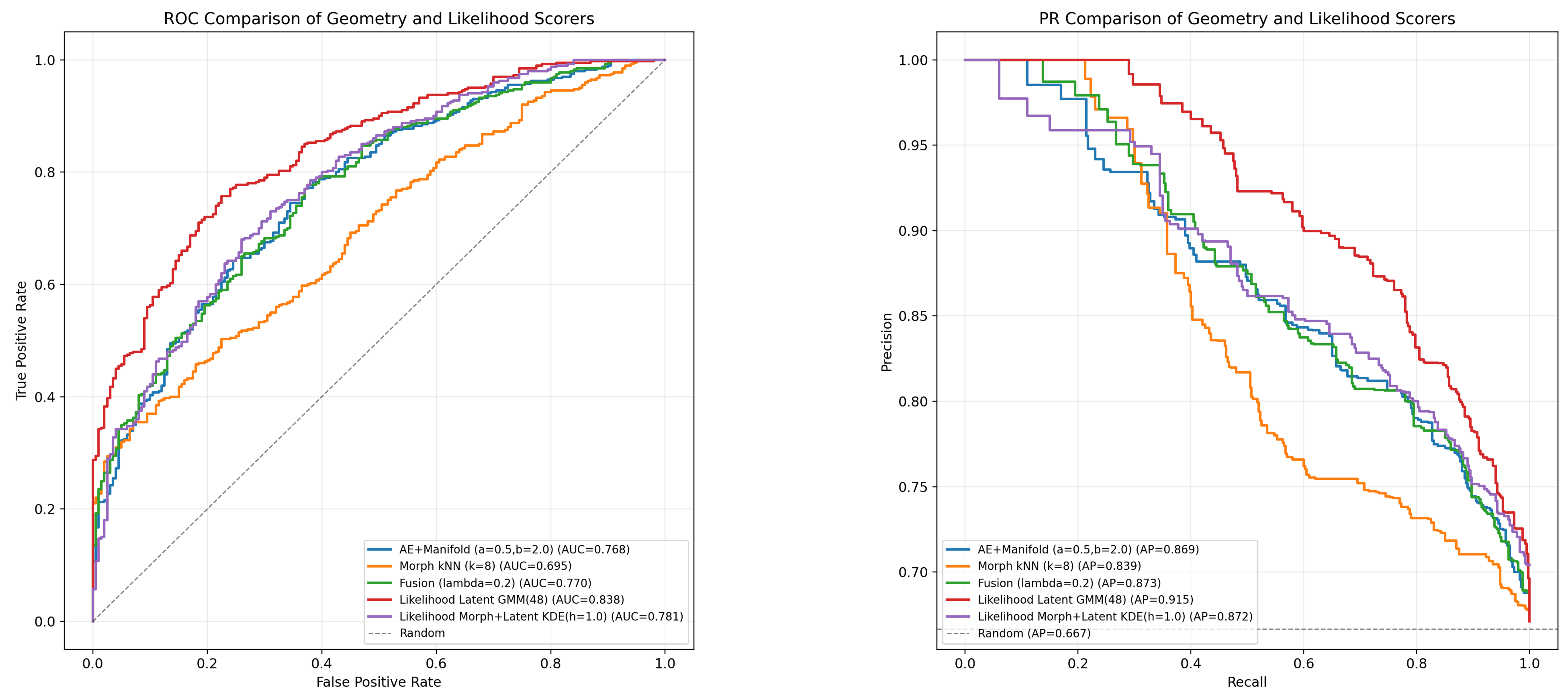


Figure 1: Representative single-run ranking curves across methods. Left: ROC (TPR vs FPR). Right:precision-recall. Legend values correspond to threshold free summary scores for that run.

HYPERPARAMETER ABLATION

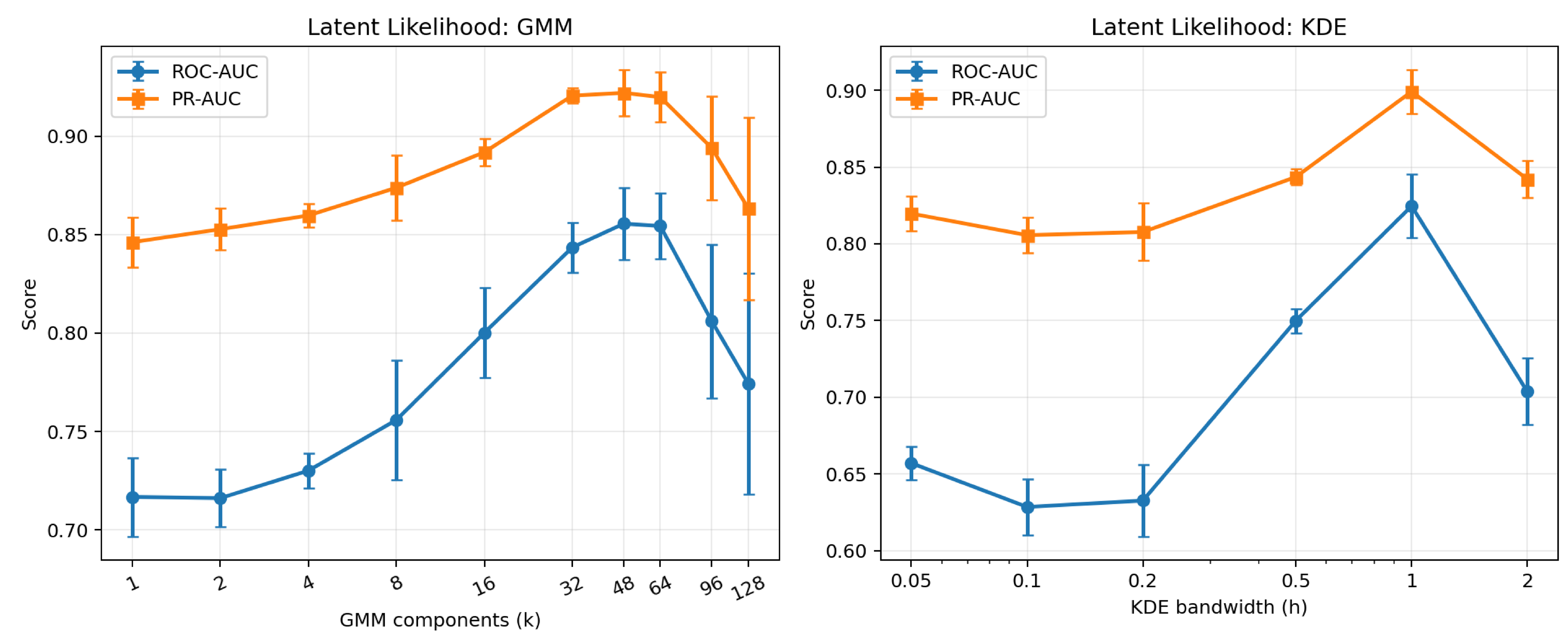


Figure 2: Latent-likelihood ablation. Left: ROC-AUC and PR-AUC versus GMM component count. Right: ROC-AUC and PR-AUC versus KDE bandwidth (log-scaled). Values are mean $\pm$ std across three seeds.

DISCUSSION

Likelihood-based scoring consistently outperforms local manifold-distance methods under the fixed latent representation studied here, suggesting that the confusion foreground contains global distributional structure not fully captured by nearest-neighbor geometry alone. Future work will examine whether these scoring methods remain stable under invertible transformations of the latent coordinates, including rotations, whitening, and nonlinear reparameterizations of the latent space.

REFERENCES

[1] LISA Collaboration. Laser interferometer space antenna, 2017. arXiv:1702.00786.

[2] Jericho Cain. Manifold learning for source separation in confusion-limited gravitational-wave data, 2025. arXiv:2511.12845, Under minor revision at Classical and Quantum Gravity.

[3] Jericho Cain. Likelihood-based one-class scoring in cwt latent space for confusion-limited lisa gravitational-wave detection, 2026. arXiv:2602.20212.